Social Networking Friend Recommendation Using Graphs

Introduction

Social networking websites have become an important part of modern communication, allowing people to connect and interact with friends, relatives, and other users across the world. As the number of users on these platforms increases, it becomes difficult for individuals to discover and connect with new people who may share common relationships or interests. A friend recommendation system helps solve this problem by analyzing the existing network of connections and suggesting potential friends to users. This project focuses on developing a simple Social Networking Friend Recommendation System using the concept of Graph Theory.

Objective

The main objective of this project is to design and implement a system that recommends new friends to users by analyzing mutual connections within a social network. The system aims to provide accurate and relevant suggestions by identifying users who share the highest number of mutual friends with the selected user.

Methodology

In this project, the social network is represented using a graph data structure. Each user is represented as a node (vertex), and friendship between users is represented as an edge connecting those nodes. The system analyzes connections using a mutual friend recommendation algorithm. It finds friends of existing friends, removes users who are already connected, counts the mutual connections, and recommends users with the highest mutual friend count.

System Implementation

The project consists of a web-based application developed using HTML and CSS for the frontend and Java for the backend. The user selects a user profile through the interface and requests recommendations. The Java backend processes the request, applies the graph algorithm, and returns suitable friend suggestions based on mutual relationships.

Results and Outcome

The system successfully provides meaningful friend recommendations by examining the social graph. For example, if Alice is connected to Bob and Charlie, and both of them are connected to David, then David receives a higher recommendation score because he has multiple mutual connections.

Conclusion

The Social Networking Friend Recommendation System demonstrates the practical application of graph theory in social media applications. By representing users as nodes and friendships as edges, the system efficiently discovers possible connections. The project can be further improved by adding databases, user interests, location-based recommendations, and advanced machine learning techniques to support larger networks.